Samyak Chakrabarty

PhD in Electrical and Computer Engineering

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Education

University of California, Los Angeles 3.92/4

Ms-PhD, Electrical and Computer Engineering, Advisor: Prof. Sudhakar Pamarti *2023 - Present*

Indian Institute of Technology, Kharagpur, India 9.39/10

B.Tech, Electronics and Electrical Communication Engineering *2019 - 23*

Projects

- **NVM based Rewritable CiM using per-cell CTT DAC** **Oct'25 - Feb'26**

  Prof. Sudhakar Pamarti, University of California, Los Angeles (Master's Thesis)

  – Proposed a novel DRAM-based bit-cell architecture improving compute density, accuracy, and energy efficiency.
  – Designed and implemented circuit schematics and layouts, performing extensive Cadence Virtuoso analog simulations to verify power, area, and linearity across process corners.
  – Co-developed a system-level integration flow enabling full-network verification of accuracy, throughput, and power performance. Compared various trade-offs between system level and hardware choices.

- **Intuitive Design and Analysis Method for MPROs** **May'22 - Aug'22**

  Prof. Sudip Shekhar, University of British Columbia, Vancouver (Summer Internship)

  – Simulated Multipath Ring Oscillators (MPROs) on Virtuoso Cadence and defined an algorithm to classify them using parameters like ring and wiring modes, successfully identifying multiple prior structures as MPROs.
  – Performed extensive time-domain, frequency-domain, phase-noise and mismatch analyses on Virtuoso Cadence.
  – Validated the proposed classification and design methodology through post-layout simulations and silicon measurements on a TSMC 65nm test chip, confirming frequency stability and mode behavior.

- **Vt-Degradation Effect of Charge Trap Flash & Impact on ANN Training** **May'21 - Dec'21**

  Prof. Udayan Ganguly, Indian Institute of Technology, Bombay (Internship)

  – Developed mathematical model of Charge Trap Flash for RPU on MATLAB, based on experimental data, to predict behaviour when subjected to stochastic pulse train.
  – Analyzed the effects of Vt-degradation and other non-idealities of CTF on RPU training using the model, to determine optimum pulse width and height.
  – Designed SPICE model of CTF transistor and its programming/erasing mechanism using DC components like VCCS and capacitors incorporating the Vt-degradation effect.

Publications

- **S. Chakrabarty**, V. Jacob, M. Zeinali, G. Karfakis, S. Qiao, Z. Guo, P. Gupta, S. Iyer, S. Pamarti. *PERCEL: A Rewritable NVM CIM Incorporating a CTT-Based Per-Cell DAC*. **IEEE JxCDC'26** | Link

- S. Shrivastava, V. Saraswat, G. Dash, **S. Chakrabarty**, and U. Ganguly. *Non-ideal program-time conservation in charge trap flash for deep learning*. **SST'23** | Link

Awards and Achievements

- **Qualcomm Innovation Fellowship North America**, 600+ Participants *Jun'26*
- **Best Masters Thesis (Gold)**, ECE, UCLA Graduating Batch *May'26*
- **Best Poster, IBM AI Symposium 2025**, Among 150+ Student posters *Nov'25*
- **Department Fellowship**, Recipient among 300+ students at ECE, UCLA. *Apr'23*
- **Best B.Tech Project Work**, Recipient among 200+ students in ECE, IIT Kharagpur. *Sep'23*
- **MITACS**, Funded Research Fellowship in University of British Columbia, Vancouver *Mar'22*